

Euclid's Elements — Book V

Proposition 8

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

Of unequal magnitudes, the greater has to the same a greater ratio than the less has; and the same has to the less a greater ratio than it has to the greater.

1 Introduction

Proposition V.8 is the first proposition in the present reconstruction of Book V where the Eudoxean comparison of ratios requires a genuinely Archimedean argument. The statement is elementary to read but structurally important: if one positive magnitude is greater than another, then this strict inequality is detected by a suitable pair of positive multiples in the sense of Euclid's Definition 7.

The proposition is also the first substantial consumer of the general magnitude layer developed in `HilbertInterfaceV.lean`. The formal proof uses strict order on positive segment classes, positive differences, iterated addition, Archimedes' principle, and a least-exceeding-multiple construction.

The production file is

```
CGJteamLab/Proposition5_8.lean
```

and it imports only

```
CGJteamLab.HilbertInterfaceV
```

from the project-level Book V infrastructure.

2 Euclid's Statement

Euclid states Proposition V.8 as follows:

Of unequal magnitudes, the greater has to the same a greater ratio than the less has; and the same has to the less a greater ratio than it has to the greater.

In the present positive-magnitude notation, if $c < a$, the proposition has two parts:

$$a : d > c : d,$$

and

$$d : c > d : a.$$

The first compares two ratios with the same second term. The second compares two ratios with the same first term.

3 Magnitude Configuration

There is no planar or spatial diagram in V.8. The configuration consists of positive magnitudes and their positive integer multiples.

For the first half, assume

$$c < a.$$

By positive difference there exists a positive magnitude e such that

$$c + e = a.$$

The formal proof then chooses a multiple of e large enough to exceed the common second magnitude d :

$$d < M_m(e).$$

It next chooses the least n such that

$$M_m(c) < M_n(d).$$

The comparison at this least index is the decisive Eudoxean witness.

For the second half, the same structure is used after writing

$$c + e = a$$

with the roles of first and second terms interchanged.

4 Eudoxus Definition 5 and Definition 7

The reconstruction distinguishes two notions.

4.1 Equality of ratios

The relation

```
HilbertEudoxusProportion Geo a b c d
```

formalizes Euclid's Definition 5. For every pair of positive natural multiple indices m, n , the three possible comparisons

$$M_m(a) < M_n(b), \quad M_m(a) = M_n(b), \quad M_n(b) < M_m(a)$$

must agree with the corresponding comparisons between c and d .

4.2 Greater ratio

The relation

```
HilbertEudoxusGreaterRatio Geo a b c d
```

formalizes the strict comparison of ratios used in V.8. It requires positive-multiple indices m, n such that

$$M_n(b) < M_m(a)$$

while

$$\neg(M_n(d) < M_m(c)).$$

The negation in the second clause is essential. Equality on the $c : d$ side is allowed; replacing this clause by the strict opposite inequality would be stronger than the Eudoxean definition used here.

5 Positive Magnitude Infrastructure

The objects of the proof are

```
HilbertPositiveSegmentClass Geo
```

rather than raw point pairs. A positive segment class is a congruence class of non-null Hilbert segments.

The strict order is

```
HilbertPositiveSegmentLess Geo a b
```

and is induced from the geometric strict segment order.

The proof also uses positive addition and positive natural multiples:

```
hilbertPositiveSegmentMultiple Geo n a
```

with shifted indexing:

$$M_0(a) = a, \quad M_1(a) = a + a, \quad M_2(a) = a + a + a, \quad \dots$$

Thus every natural index already denotes a positive multiple; there is no zero magnitude in the type.

6 The Archimedean Step

The central infrastructure theorem is

```
hilbertPositiveSegment_archimedean
```

which states that for positive magnitudes a, b there exists m such that

$$b < M_m(a).$$

This theorem is not an additional axiom of Book V. It is derived in `HilbertInterfaceV.lean` from the existing geometric Archimedean structure

```
HilbertArchimedeanPlane
```

after normalizing the weak Hilbert segment chain to positive segment classes.

For V.8 the theorem is applied to the positive remainder e . In the first half it gives

$$d < M_m(e).$$

This is the step that makes it possible to separate the two ratios by finite multiples.

7 Least Exceeding Multiple

Archimedes gives existence of an exceeding multiple, but V.8 also needs minimality.

The reusable theorem is

```
hilbertPositiveSegment_exists_least_exceeding_multiple
```

and produces a least n such that

$$x < M_n(a),$$

together with

$$k < n \implies \neg(x < M_k(a)).$$

For the first half of V.8 we apply it with

$$x = M_m(c), \quad a = d.$$

Hence

$$M_m(c) < M_n(d),$$

and no earlier multiple of d already exceeds $M_m(c)$.

This least-index condition is the formal replacement for the implicit “first multiple beyond” step in the classical magnitude argument.

8 Euclid’s Classical Proof Pattern

The first half of the proposition can be organized as follows.

Assume

$$c < a.$$

Take the positive difference e with

$$a = c + e.$$

Choose m such that

$$d < M_m(e).$$

Choose the least n such that

$$M_m(c) < M_n(d).$$

By minimality, the preceding multiple does not exceed $M_m(c)$. Consequently the next step of size d , together with the inequality $d < M_m(e)$, forces

$$M_n(d) < M_m(c) + M_m(e).$$

Since positive multiplication distributes over positive addition,

$$M_m(c) + M_m(e) = M_m(c + e) = M_m(a).$$

Therefore

$$M_n(d) < M_m(a).$$

At the same time,

$$M_m(c) < M_n(d),$$

so by asymmetry

$$\neg(M_n(d) < M_m(c)).$$

The pair m, n therefore witnesses

$$a : d > c : d.$$

The second half is the dual argument with the common first term fixed.

9 Classical Dependency Pattern

The mathematical dependency pattern used in the reconstruction is

$$\begin{array}{c}
c < a \\
\Downarrow \\
\exists e > 0, c + e = a \\
\Downarrow \\
\text{Archimedes: choose } m \text{ with } d < M_m(e) \\
\Downarrow \\
\text{least } n \text{ with } M_m(c) < M_n(d) \\
\Downarrow \\
M_n(d) < M_m(a) \\
\Downarrow \\
\neg(M_n(d) < M_m(c)) \\
\Downarrow \\
a : d > c : d.
\end{array}$$

The second clause of V.8 follows from the same architecture after interchanging the two ratio positions.

10 Formal Statement

The production theorem is:

```

theorem euclid_proposition_5_8
  [HilbertIncidence Geo]
  [HilbertArchimedeanPlane Geo]
  (a c d : HilbertPositiveSegmentClass Geo)
  (hca : HilbertPositiveSegmentLess Geo c a) :
  HilbertEudoxusGreaterRatio Geo a d c d /\
  HilbertEudoxusGreaterRatio Geo d c d a := by

constructor
case left =>
  exact
    hilbertEudoxus_v8_greater_first_same_second
      Geo a c d hca
case right =>
  exact
    hilbertEudoxus_v8_same_first_lesser_second
      Geo d c a hca

```

The final Euclid-numbered theorem is intentionally short. The substantive proof is carried by the two proposition-local lemmas:

```

hilbertEudoxus_v8_greater_first_same_second
hilbertEudoxus_v8_same_first_lesser_second

```

which formalize the two halves of Euclid’s statement.

11 Lean Proof Architecture: First Half

The first proposition-local theorem is

```

hilbertEudoxus_v8_greater_first_same_second

```

with target

$$c < a \implies a : d > c : d.$$

The proof proceeds through the following data.

First,

```

hilbertPositiveSegment_exists_add_of_less

```

produces e with

$$c + e = a.$$

Second,

`hilbertPositiveSegment_archimedean`

produces m with

$$d < M_m(e).$$

Third,

`hilbertPositiveSegment_exists_least_exceeding_multiple`

produces the least n such that

$$M_m(c) < M_n(d).$$

The proof then splits on n .

11.1 Case $n = 0$

Here

$$M_0(d) = d.$$

Since

$$d < M_m(e)$$

and

$$M_m(e) < M_m(c) + M_m(e),$$

we obtain

$$d < M_m(c) + M_m(e) = M_m(a).$$

The least-index inequality itself gives

$$M_m(c) < d,$$

hence asymmetry supplies the required negated comparison.

11.2 Case $n = k + 1$

Minimality gives

$$\neg(M_m(c) < M_k(d)).$$

By trichotomy either

$$M_k(d) < M_m(c)$$

or

$$M_k(d) = M_m(c).$$

The reverse strict inequality is impossible by minimality.

Combining either remaining case with

$$d < M_m(e)$$

gives

$$M_k(d) + d < M_m(c) + M_m(e).$$

Using the recursion for positive multiples and distributivity,

$$M_{k+1}(d) < M_m(a).$$

Thus m, n satisfy Definition 7.

12 Lean Proof Architecture: Second Half

The second proposition-local theorem is

```
hilbertEudoxus_v8_same_first_lesser_second
```

with target

$$b < d \implies a : b > a : d.$$

The proof is dual.

Write

$$b + e = d.$$

Choose n such that

$$a < M_n(e),$$

then choose the least m such that

$$M_n(b) < M_m(a).$$

The minimality analysis of m shows

$$M_m(a) < M_n(d).$$

Hence asymmetry gives

$$\neg(M_n(d) < M_m(a)),$$

while the least-exceeding construction already gives

$$M_n(b) < M_m(a).$$

These are exactly the two clauses required to witness

$$a : b > a : d.$$

13 Hidden Formal Data

Several details that are implicit in Euclid’s magnitude language must be explicit in Lean.

13.1 All magnitudes are positive

The type

```
HilbertPositiveSegmentClass Geo
```

contains only non-null segment classes. Therefore subtraction is not a primitive total operation. The positive difference e is obtained only under a strict inequality.

13.2 Positive multiples use shifted indices

The index 0 denotes one copy, not zero copies. This is why the least exceeding multiple proof has a genuine base case $n = 0$.

13.3 Minimality is explicit

The phrase “take the first multiple which exceeds” is formalized as an existence theorem over $\mathbb{N}$ with a universal minimality clause.

13.4 The greater-ratio witness is existential

V.8 does not prove a numerical quotient inequality. It produces the multiple indices m, n required by Eudoxus Definition 7.

14 Relationship with Earlier and Later Book V Propositions

In the present reconstruction V.8 is the first Euclid-numbered theorem in Book V that uses the full Archimedean magnitude infrastructure.

Its main later consumers are V.14 and, through V.14, V.16.

For equal ratios

$$a : b = c : d,$$

V.14 needs to transport strict comparison from a, c to b, d . The strict part begins with V.8: from $a < c$ one obtains a strict comparison of ratios with a common second term. V.13 transports that ratio comparison through the given equality, and V.10 converts it back to an inequality of magnitudes.

Thus the key dependency chain is

$$V.8 \longrightarrow V.13 \longrightarrow V.10 \longrightarrow V.14 \longrightarrow V.16.$$

This is also why V.8 was the first substantial theorem formalized when the Book V layer was introduced for the later reconstruction of XI.23.

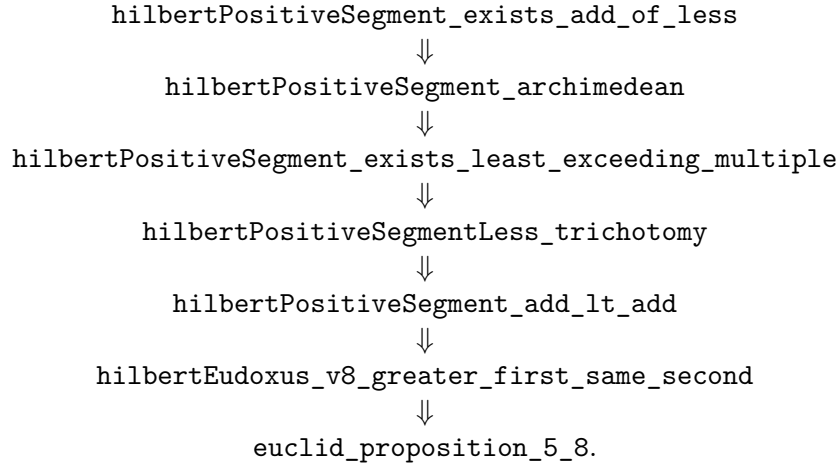
15 Classical DAG versus Formal DAG

The formal proof keeps the classical Eudoxean structure but makes several implicit steps explicit.

Magnitude-level DAG.

$$\begin{array}{c}
 c < a \\
 \Downarrow \\
 c + e = a \\
 \Downarrow \\
 d < M_m(e) \\
 \Downarrow \\
 \text{least } n : M_m(c) < M_n(d) \\
 \Downarrow \\
 M_n(d) < M_m(c) + M_m(e) \\
 \Downarrow \\
 M_n(d) < M_m(a) \\
 \Downarrow \\
 a : d > c : d.
 \end{array}$$

Lean-level DAG.



The second half has the same architecture with the ratio positions interchanged.

16 Structural Lesson

V.8 shows why Book V cannot be reduced to a few symbolic ratio identities. The strict comparison of ratios depends on the order theory of magnitudes and on an Archimedean finite-multiple principle.

The formal reconstruction makes this especially clear. No real numbers, coordinates, lengths as scalars, or division operation are introduced. Everything is expressed synthetically in terms of:

- congruence classes of positive segments;
- strict segment order;
- positive addition;
- iterated addition;
- Archimedean existence of a sufficiently large multiple;
- well-ordering of the natural-number multiple index.

This is precisely the kind of infrastructure needed for a source-faithful Eudoxean development.

17 Conclusion

Proposition V.8 is the first major order theorem of the reconstructed Eudoxean ratio calculus.

Starting from a strict inequality of positive magnitudes, the proof constructs finite multiples that witness a strict inequality of ratios in the exact sense of Definition 7. The decisive ingredients are the positive remainder, Archimedes' principle, and the least multiple crossing a prescribed magnitude.

The final Euclid-numbered theorem is a short wrapper, but the mathematical content is substantial. It is the gateway from the additive ordered magnitude layer to the later comparison theorems

V.10, V.13, V.14, and ultimately to V.16 alternando, which is required in the classical route to Euclid XI.23.